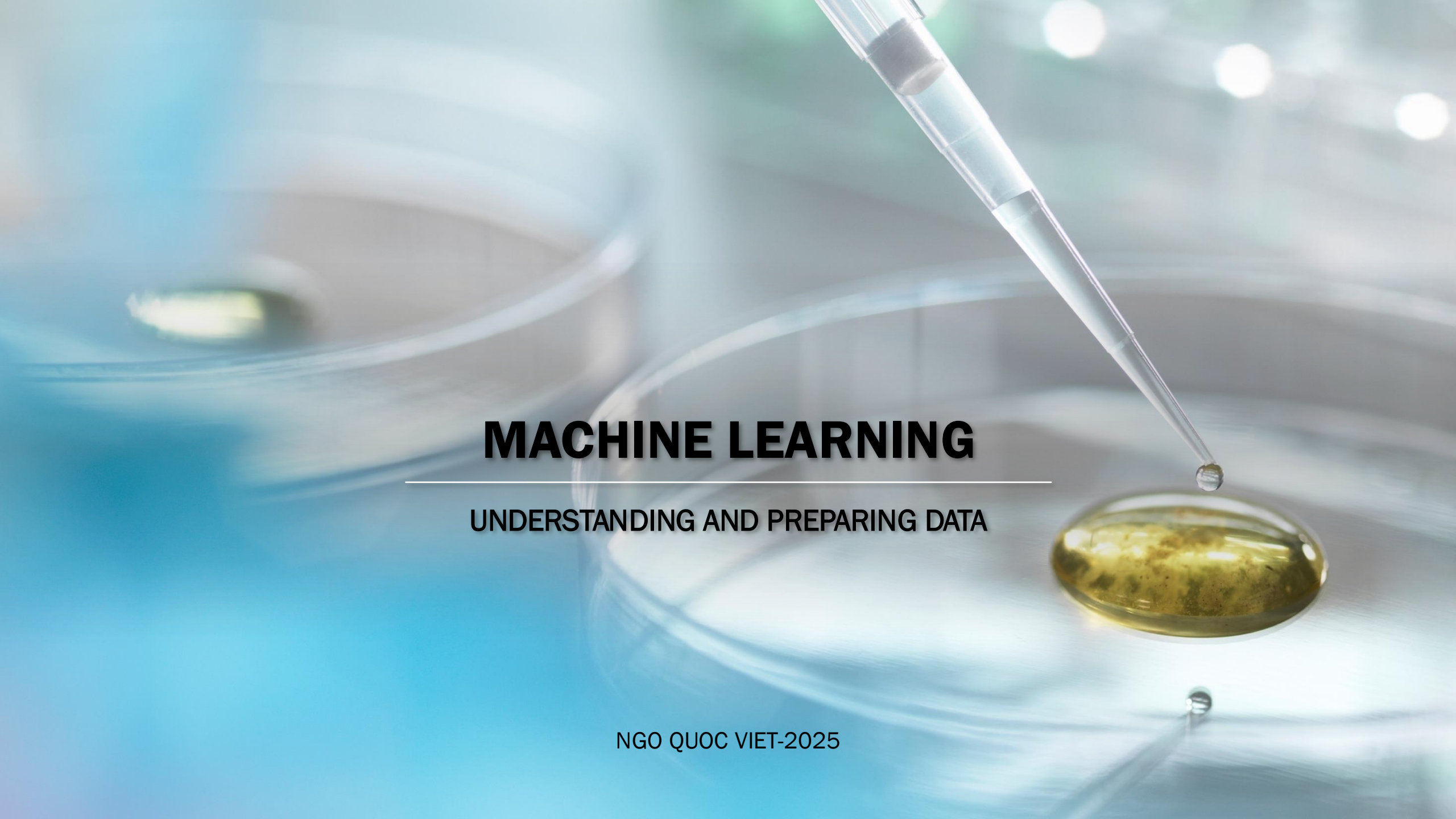


MACHINE LEARNING

UNDERSTANDING AND PREPARING DATA

NGO QUOC VIET-2025

CONTENTS


Bias-variance trade off.


Data Visualization.


Data Normalization.


Data Regularization.


Split Training and Test Sets.

BIAS & VARIANCE

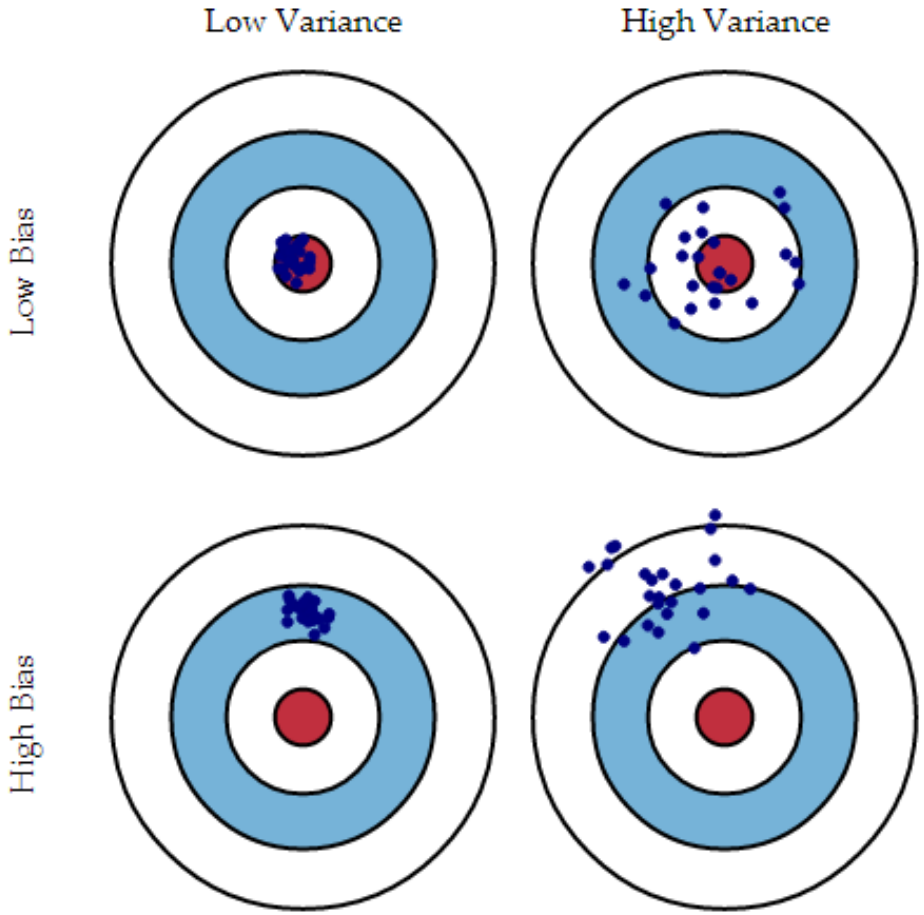
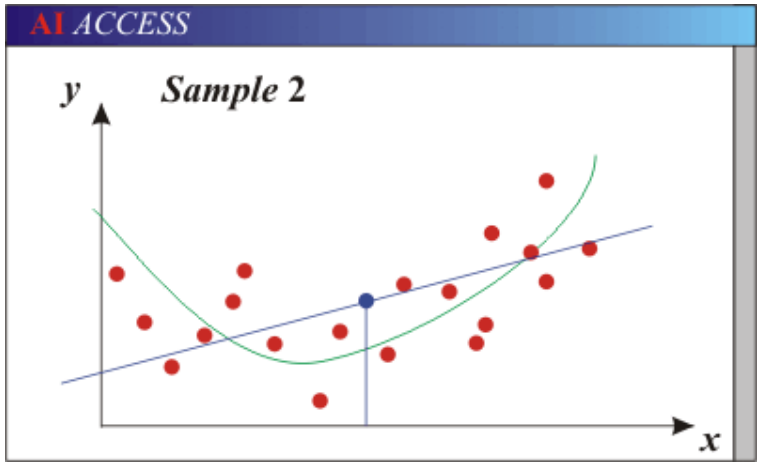
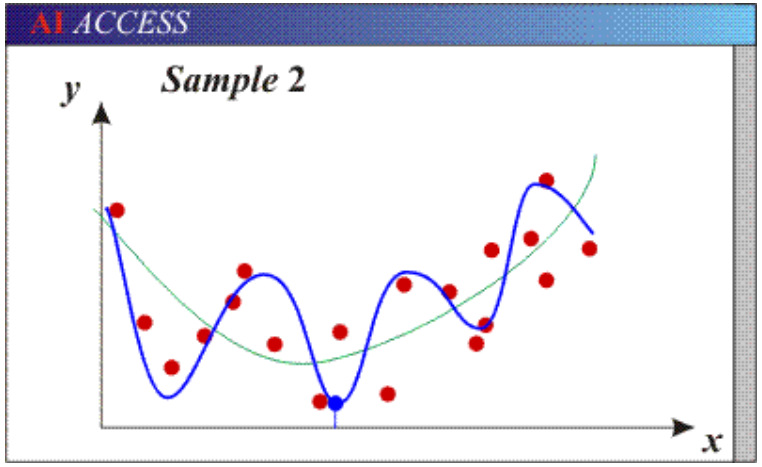


Fig. 1 Graphical illustration of bias and variance.

BIAS-VARIANCE TRADE-OFF



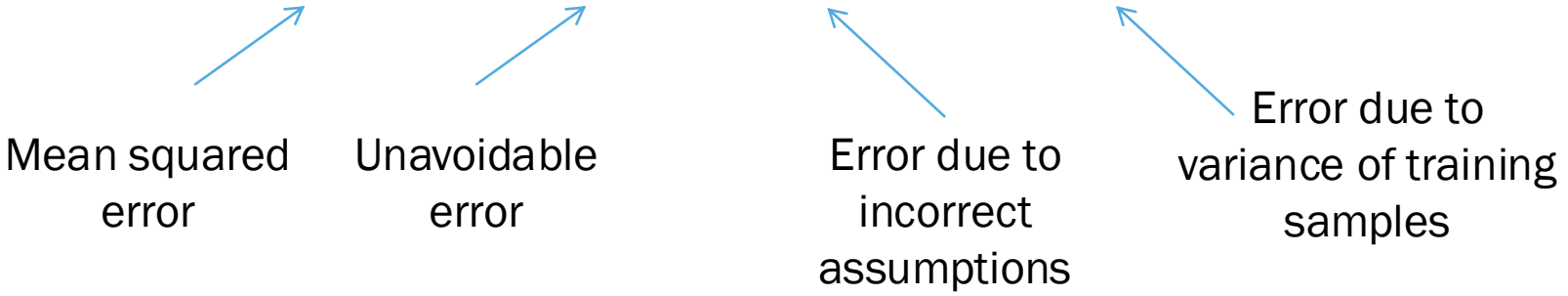
- Model is too simple, not accurate because of large bias.



- Models with too many parameters are also not accurate because of large variance (too much learning, including noise overfitting).

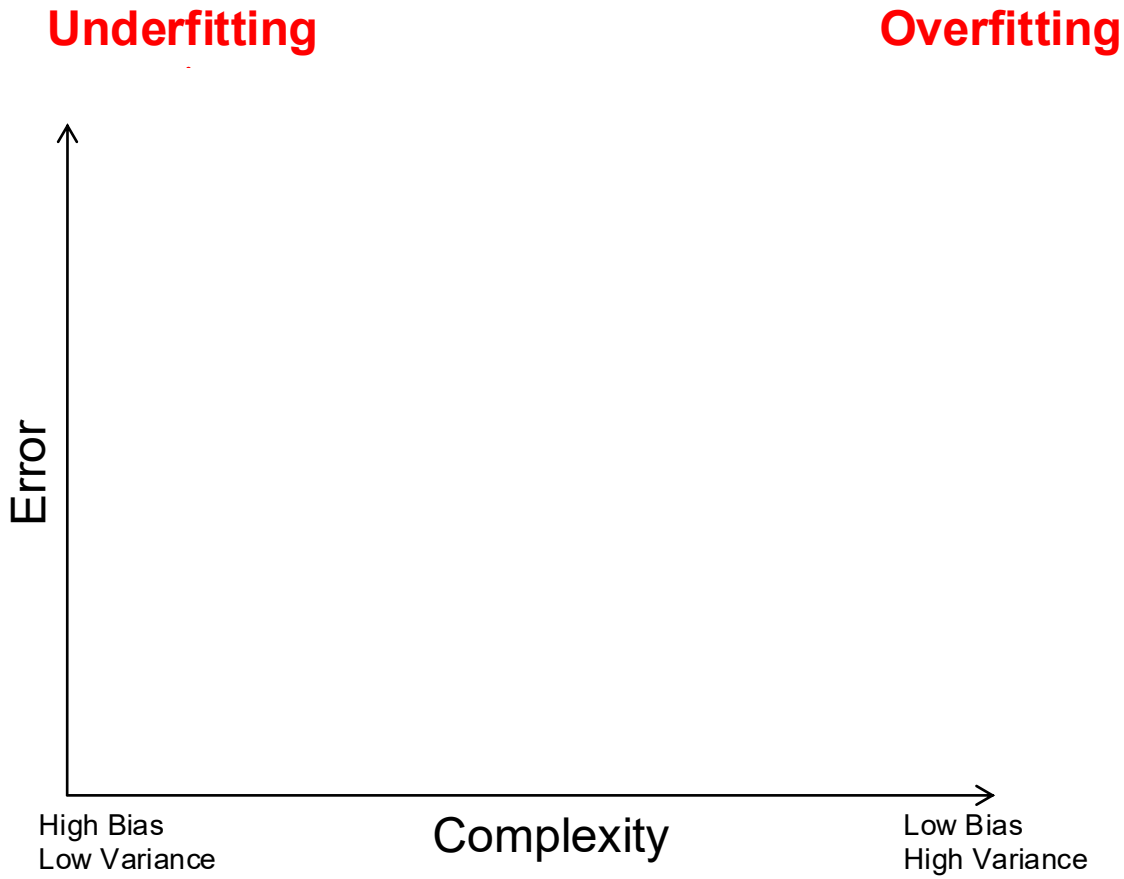
BIAS-VARIANCE TRADE-OFF

$$E(MSE) = noise^2 + bias^2 + variance$$

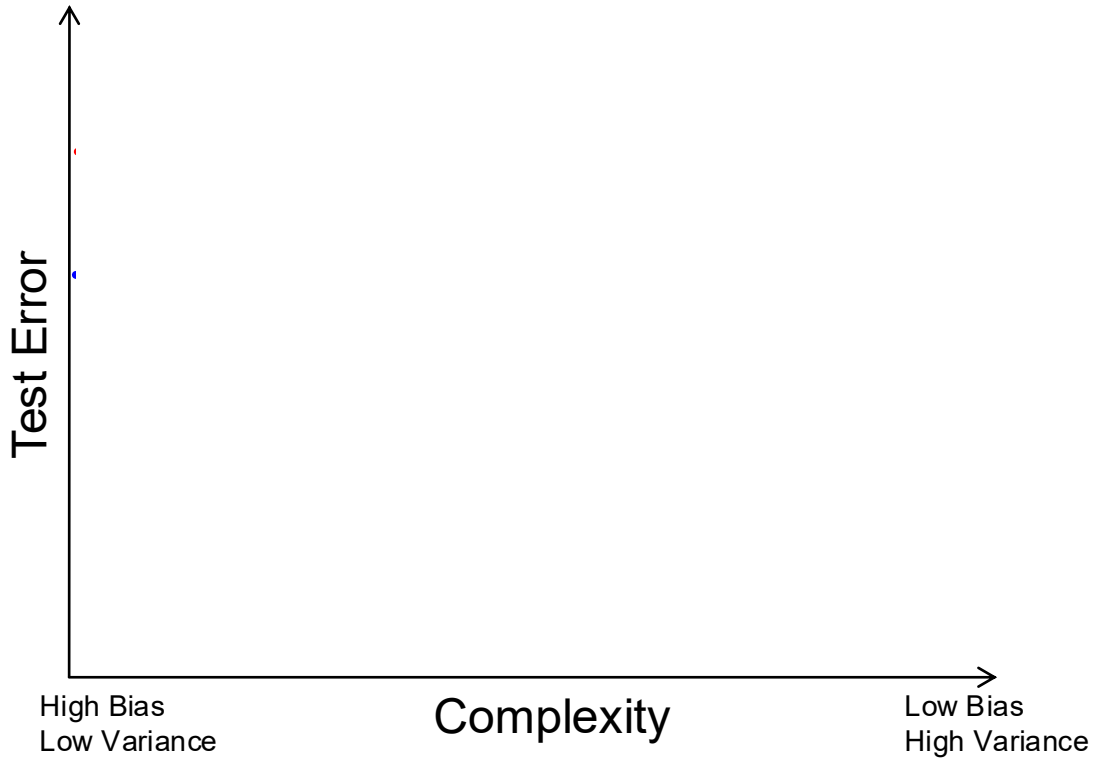


- http://scott.fortmann-roe.com/docs/BiasVariance.html
- http://web.engr.oregonstate.edu/~tgd/classes/534/slides/part9.pdf
- http://www.aiaccess.net/English/Glossaries/GlosMod/e_gm_bias_variance.htm

BIAS-VARIANCE TRADE-OFF



BIAS-VARIANCE TRADE-OFF

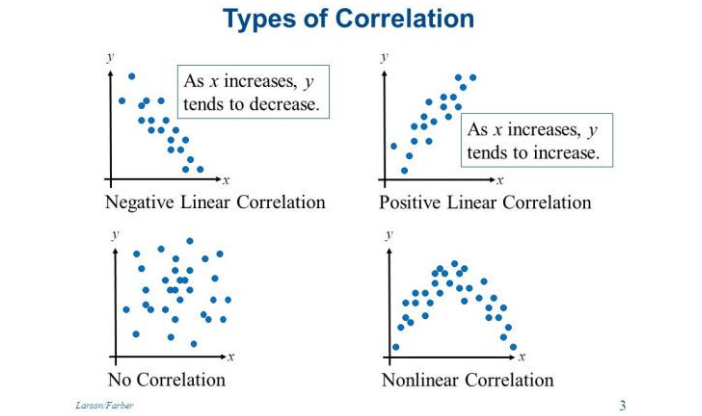


WHY DATA VISUALIZATION IS IMPORTANT?

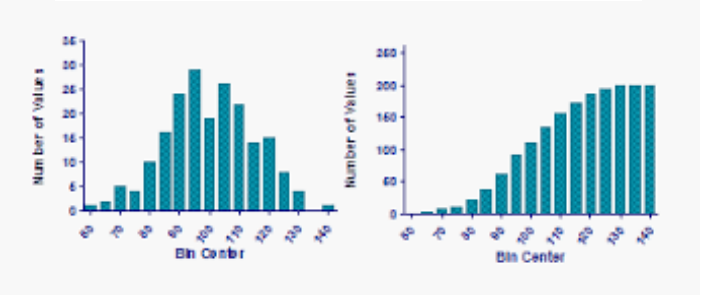
- The way the human brain processes information.
- Using charts or graphs to visualize large amounts of complex data is easier than poring over spreadsheets or reports
- To communicate information in a manner that is universal, fast, and effective.

THE BENEFITS OF DATA VISUALIZATION

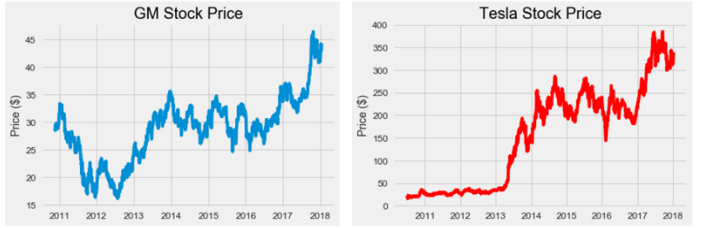
- Correlations in Relationships: identify the correlations between the relationship of independent variables.



- Frequency: The rate, or how often. Closely related to trends over time.



- Trends Over Time: It's impossible to make predictions without having the necessary information from the past and present.



DATA VISUALIZATION TECHNIQUES

- Infographics
- Heatmap
- Fever Charts
- Area Chart (or Graph)
- Histogram

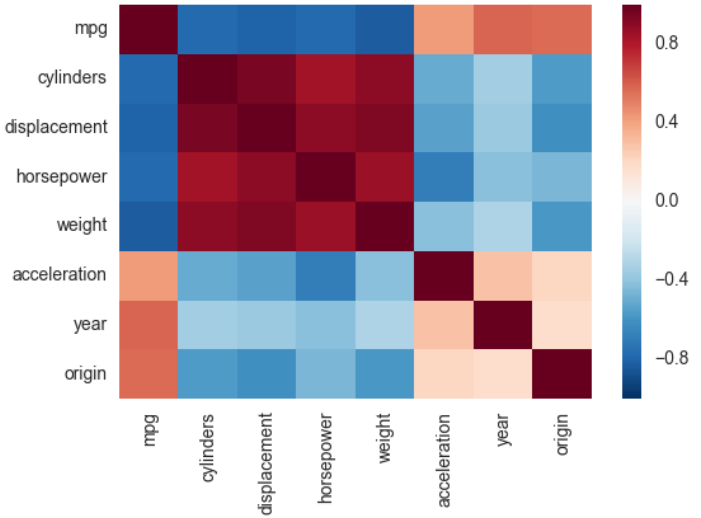
DATA VISUALIZATION: HEATMAP

- Observe the correlation between the features.

Crazy egg chart (measure the components used a lot in website, app).



Correlation coefficient chart (correlation matrix required)



https://stackoverflow.com/questions/39409866/correlation-heatmap

DATA VISUALIZATION: FEVER CHARTS

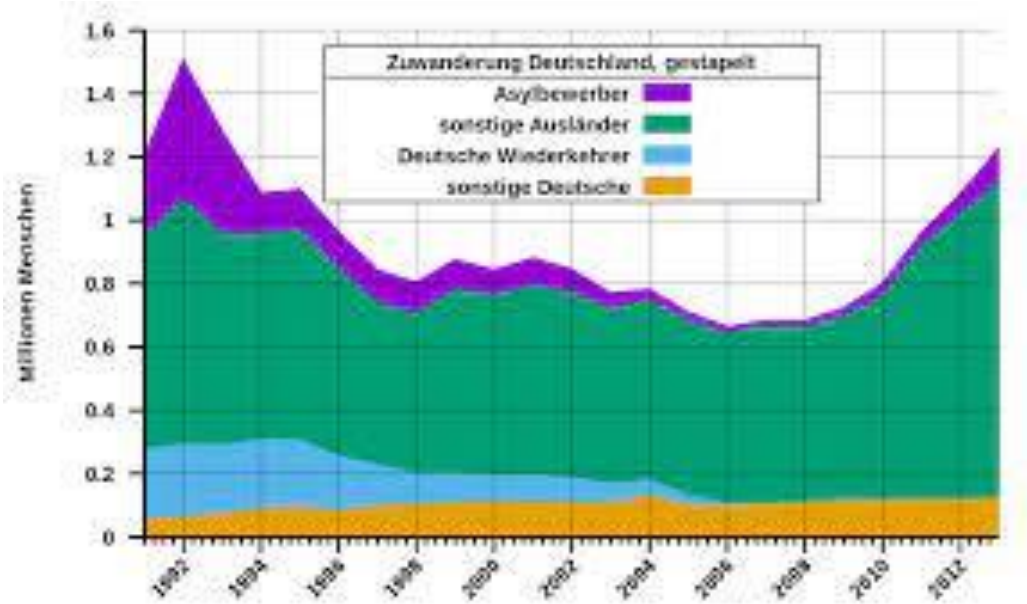
- A fever chart is a graphical representation showing change of a variable over time.
- Fever charts are sometimes called time-series charts



WhatIs.com

DATA VISUALIZATION: AREA CHART OR GRAPHS

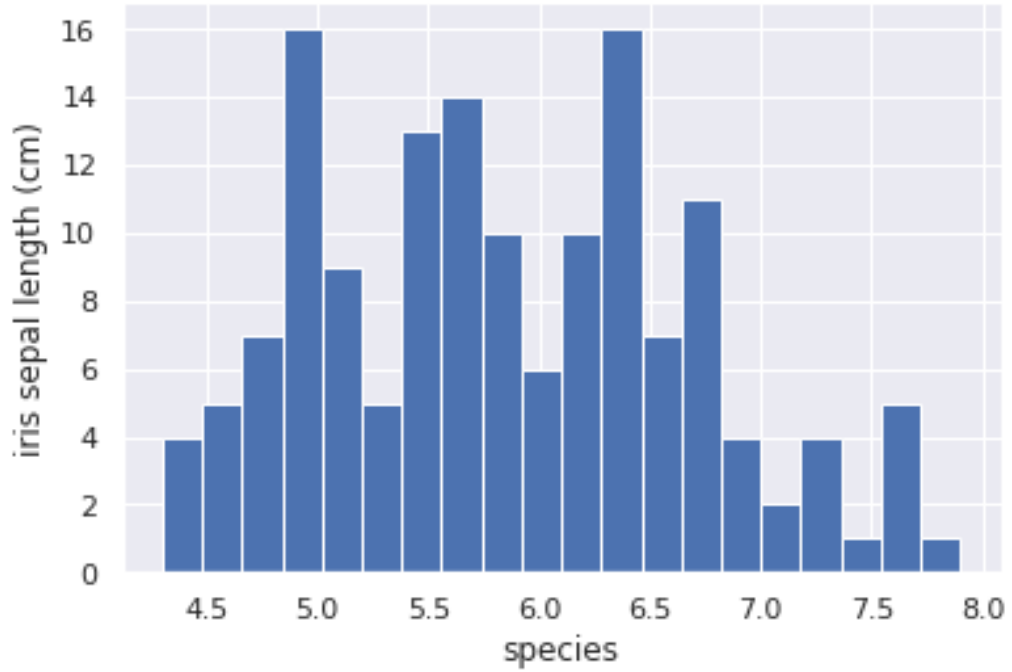
- In data visualization, an area chart is an extension of a line graph. It combines the line chart and the bar chart to reveal how one or more groups' numeric values change over a second variable.
- An area chart is a great chart to visualize a volume change over a period of time.
- Area charts are excellent when describing the overall trend.



The basic area chart is used to depict the data series as colored regions that help in comparing the values of multiple series for the same data point.

DATA VISUALIZATION: HISTOGRAM

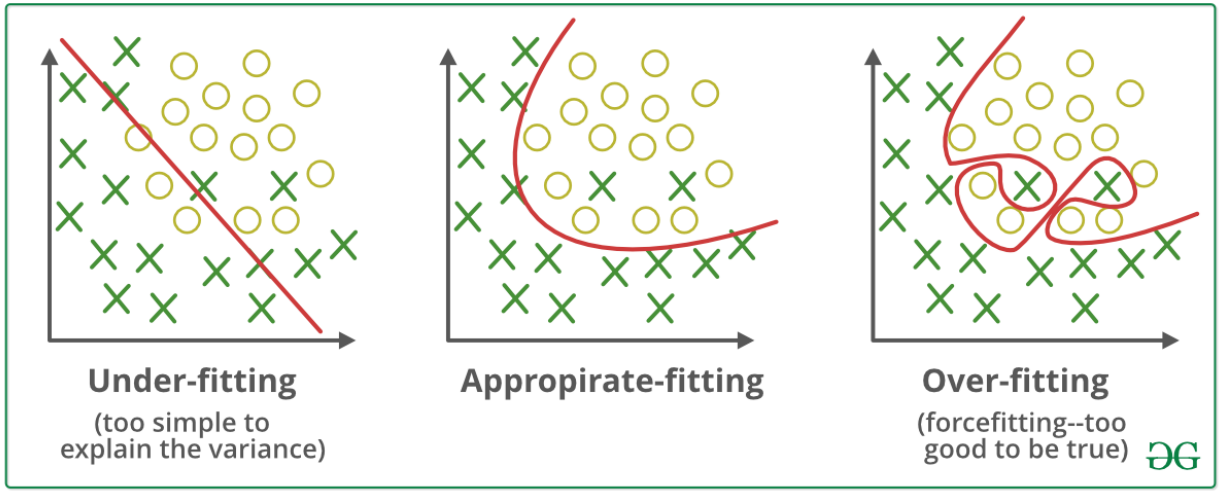
- Trực quan phân phối trong những khoảng giá trị được xác định trước của một đặc trưng.



DATA VISUALIZATION: SIMPLE CODE

- https://www.kaggle.com/jeffd23/scikit-learn-ml-from-start-to-finish
- https://realpython.com/pandas-plot-python/
- https://medium.com/data-science-bootcamp/data-visualization-in-machine-learning-beyond-the-basics-baf2cbea8989.

PROBLEMS OF FITTING



- Over fitting:
  - Building a model that matches the training data “too closely”.
  - Learning from the error/distrubance/noise in the data, rather than just the truevalues/signal.
- How does over-fitting occur?
  - Evaluating a model by testing it on the same data that was used to train it.
  - Creating a model that is “too complex”.
- Impact of over fitting: well on the training data, but won’t generalize.

DATA NORMALIZATION

- Goal of normalization is to change the values of numeric columns in the dataset to a common scale.
- For machine learning, every dataset does not require normalization. It is required only when features have different ranges.
  - Eg: Although age (0-100) and income (0-billions) are in different ranges, but it does not require normalization.
- **Normalization** is a good technique to use when you do not know the distribution of your data or when you know the distribution is not Gaussian.
- **Normalization** is useful when your data has varying scales and the algorithm you are using does not make assumptions about the distribution of your data, such as k-nearest neighbors and artificial neural networks.

DATA NORMALIZATION: DEMONSTRATION

Use ANN (3 layers) to predict forest cover. Here are few rows of original data (https://archive.ics.uci.edu/ml/datasets/covertype)

Elevation	Aspect	Slope	Horizontal_D	Vertical_Dist	Horizontal_D	Hillshade_9a	Hillshade_Nc	Hillshade_3p	Horizontal_Distance_To_Fire_Points
2596	51	3	258	0	510	221	232	148	6279
2590	56	2	212	-6	390	220	235	151	6225
2804	139	9	268	65	3180	234	238	135	6121
2785	155	18	242	118	3090	238	238	122	6211
2595	45	2	153	-1	391	220	234	150	6172
2579	132	6	300	-15	67	230	237	140	6031
2606	45	7	270	5	633	222	225	138	6256
2605	49	4	234	7	573	222	230	144	6228
2617	45	9	240	56	666	223	221	133	6244
2612	59	10	247	11	636	228	219	124	6230
2612	201	4	180	51	735	218	243	161	6222
2886	151	11	371	26	5253	234	240	136	4051
2742	134	22	150	69	3215	248	224	92	6091
2609	214	7	150	46	771	213	247	170	6211
2503	157	4	67	4	674	224	240	151	5600
2495	51	7	42	2	752	224	225	137	5576
2610	259	1	120	-1	607	216	239	161	6096
2517	72	7	85	6	595	228	227	133	5607
2504	0	4	95	5	691	214	232	156	5572

```
model = keras.Sequential([
keras.layers.Dense(64,
activation=tf.nn.relu,
input_shape=(x_train.shape
[1],)),
keras.layers.Dense(64,
activation=tf.nn.relu),
keras.layers.Dense(8,
activation= 'softmax')
])
```

Training 26 epochs

Validation accuracy of the above model is just 48.80%

DATA NORMALIZATION: DEMONSTRATION

Normalization before prediction

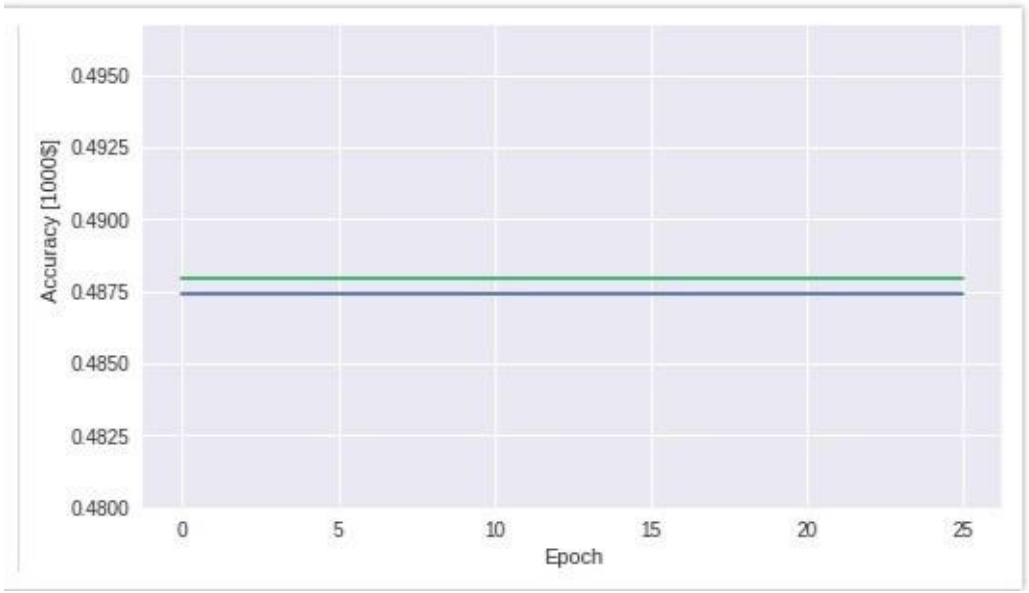
```
#Select numerical columns which needs to be normalized
train_norm = x_train[x_train.columns[0:10]]
test_norm = x_test[x_test.columns[0:10]]
# Normalize Training Data
std_scale = preprocessing.StandardScaler().fit(train_norm)
x_train_norm = std_scale.transform(train_norm)
#Converting numpy array to dataframe
training_norm_col = pd.DataFrame(x_train_norm,
index=train_norm.index, columns=train_norm.columns)
x_train.update(training_norm_col)
print (x_train.head())
# Normalize Testing Data by using mean and SD of training set
x_test_norm = std_scale.transform(test_norm)
testing_norm_col = pd.DataFrame(x_test_norm,
index=test_norm.index, columns=test_norm.columns)
x_test.update(testing_norm_col)
print (x_train.head())
```

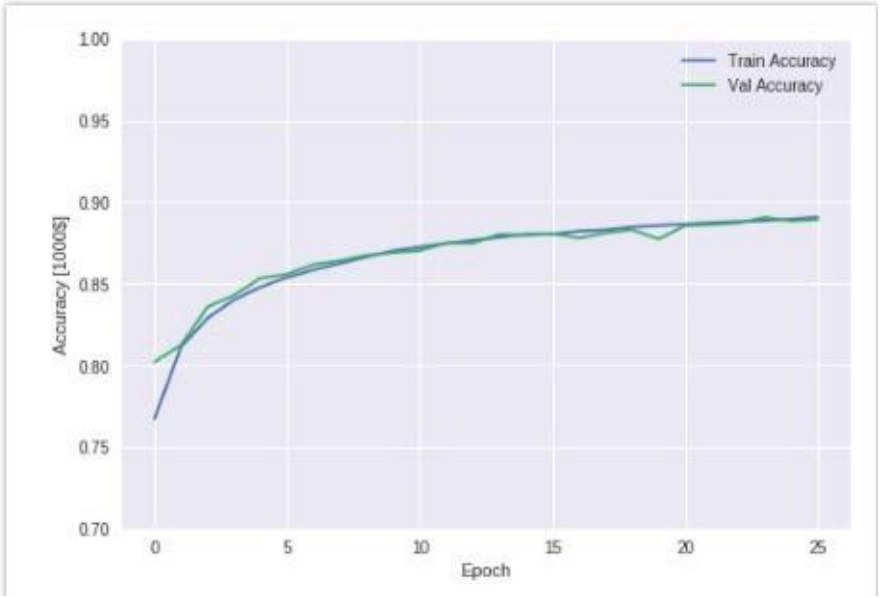
The same model, 26 training epochs.

```
model = keras.Sequential([
keras.layers.Dense(64,
activation=tf.nn.relu,
input_shape=(x_train.shape[1],)),
keras.layers.Dense(64,
activation=tf.nn.relu),
keras.layers.Dense(8, activation=
'softmax')
])
```

Validation accuracy of the same model with normalized data is 88.93%.

Left: Model Accuracy, without normalized data
Right: Model Accuracy with normalized data





Model is not able to learn in 26 epochs. It is because different features do not have similar ranges of values and hence **gradients may end up taking a long time.**

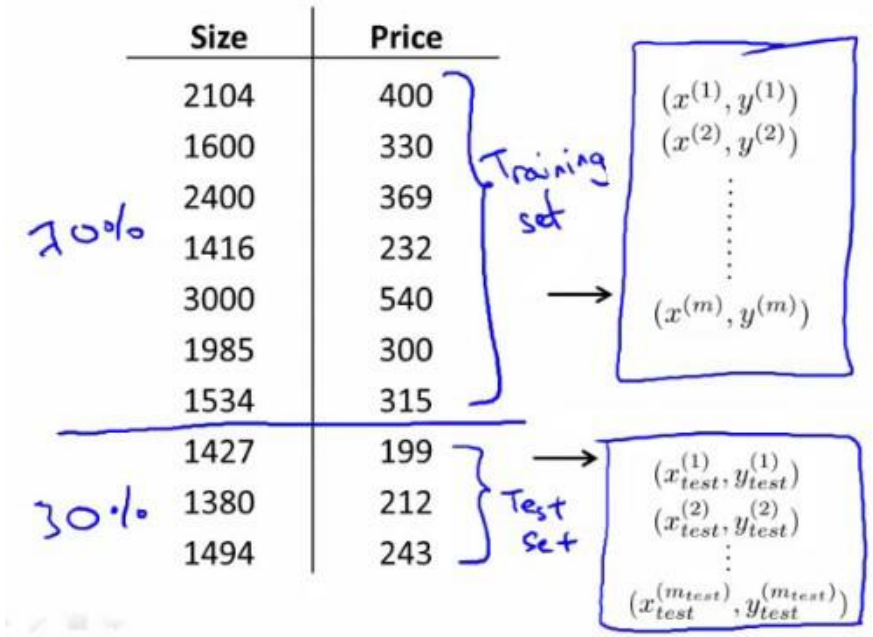
With normalizing the data, the different features take on similar ranges of values so that gradient descents can converge more quickly.

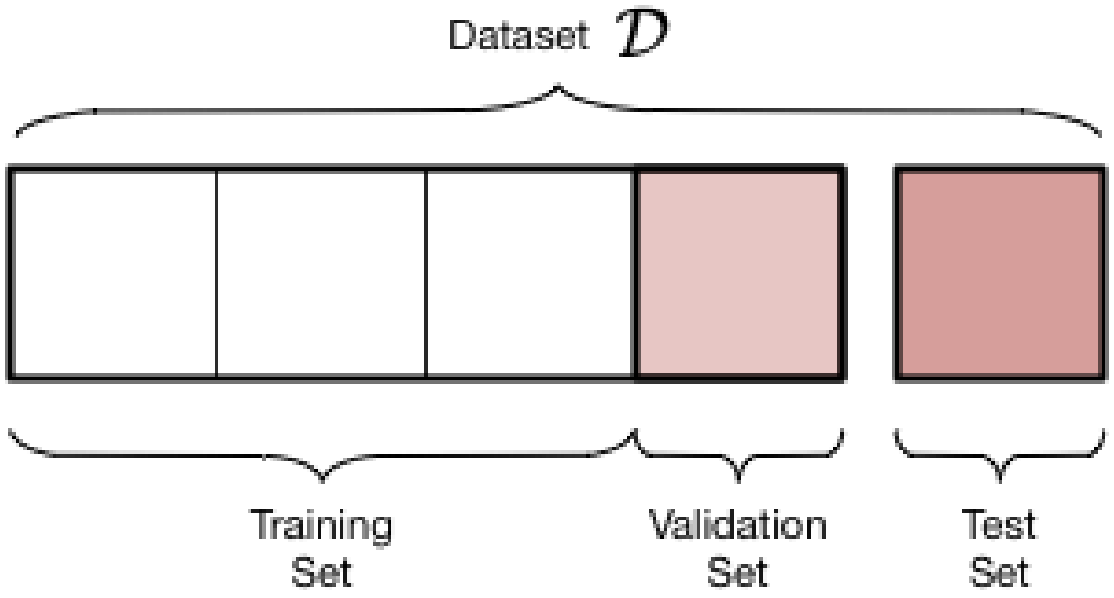
DATA NORMALIZATION TECHNIQUES

- Standard Scalar (Z-score normalization): centering the variable at zero ($\mu = 0$) and standardizing the variance at 1 ($\sigma = 1$). The minimum and maximum values vary according to how spread out the variable was, to begin with, and is highly influenced by the presence of outliers
- Min-Max Scalar: the data is scaled to a fixed range—usually 0 to 1. MinMax Scalar is sensitive to outliers.
- Robust Scalar: Scaling to median and quantiles.

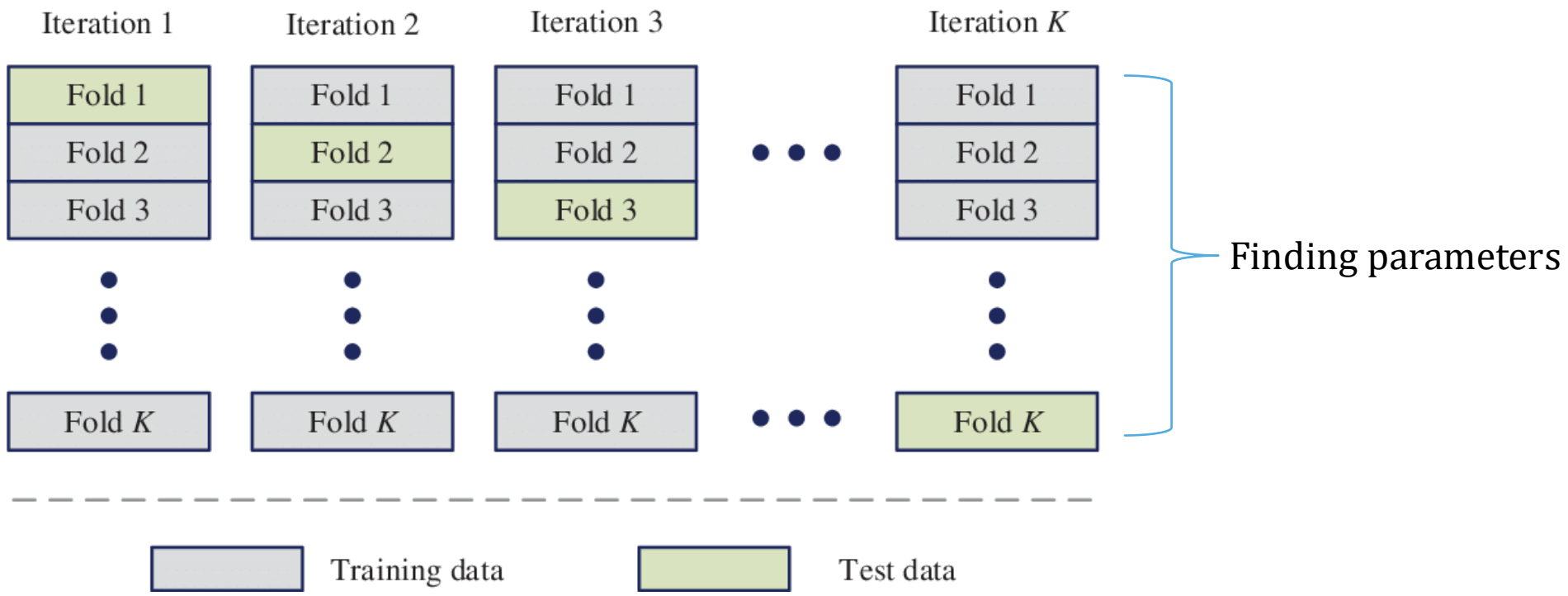
TRAIN, VALID AND TEST SPLIT

- We don't evaluate our final model performance based on training set.





K-FOLD (ON TRAINING SET) CROSS VALIDATION



To reduce overfiting we typically do stratified sampling and shuffle the data before folding

K-FOLD CROSS VALIDATION

- Model Tuning
  - Evaluate several configurations of the model.
  - Perform Grid search on hyper-parameters.
- Goal: Find the best model parameters.
- https://scikit-learn.org/stable/modules/cross_validation.html

THANK YOU